HYBRID ADAPTIVE TRANSFORMER DIFFERENTIAL PROTECTION:

AN INTELLIGENT DWT-BiLSTM FRAMEWORK WITH OPTUNA HYPERPARAMETER OPTIMISATION

by

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ABSTRACT

Power transformers constitute the most critical and capital-intensive components of electrical power transmission and distribution networks, making their reliable and rapid protection a paramount concern for ensuring continuity and stability of electric power supply. Differential protection has long been recognised as the primary scheme for safeguarding power transformers against internal faults, operating on the fundamental principle of comparing currents entering and leaving the protected zone. However, the conventional percentage differential relay faces significant challenges in accurately distinguishing between genuine internal fault conditions and non-fault transient events such as magnetising inrush currents, sympathetic inrush, and external faults with current transformer (CT) saturation. The second harmonic restraint method, widely employed to address inrush conditions, suffers from reduced sensitivity in modern transformers employing high-permeability core materials that produce second harmonic ratios as low as 7–10%, well below the typical 15–20% restraint threshold.

This thesis proposes a novel Hybrid Adaptive Transformer Differential Protection (HATDP) framework that synergistically integrates the multi-resolution feature extraction capabilities of the Discrete Wavelet Transform (DWT) with the advanced temporal classification power of a Bidirectional Long Short-Term Memory (BiLSTM) network. The proposed methodology employs a two-level Daubechies-4 (db4) wavelet decomposition on a 32-sample full-cycle sliding window at 1,600 Hz sampling rate, producing nine energy features per classification instant: detail-1 energy (400–800 Hz), detail-2 energy (200–400 Hz), and approximation-2 energy (0–200 Hz) for each of the three phases. The feature tensor is $\log_{1p}$-normalised and provided to a Bidirectional LSTM with LayerNorm and global temporal attention pooling, trained as a binary classifier (Trip / No-Trip) using PyTorch with BCEWithLogitsLoss, AdamW optimiser, and Optuna Bayesian hyperparameter optimisation (50 trials, TPE sampler). A critical innovation of this thesis is the hybrid decision handshake architecture wherein the BiLSTM classifier acts as a supervisory Veto for the classical adaptive 87T differential relay logic — the relay only trips if both the 87T element operates and the BiLSTM classifies the event as an internal fault with confidence exceeding a Youden J-optimised threshold.

A comprehensive simulation model of a 300 MVA, 230/11 kV, Yd1 power transformer is developed in MATLAB/Simulink (R2024a, Simscape Electrical).

Automated dataset generation is performed through an extended ControlPanel_StressTest.m script implementing Step1/Step2 external breaker control, covering 43 distinct scenario types: all 11 internal fault types (AG through ABCG) at both low-impedance (0.01–2 Ω) and high-impedance (50–300 Ω) levels, 9 external fault types, evolving faults, CT saturation stress, fault-during-inrush, and external fault during inrush. The dataset includes rich per-sample metadata (fault resistance, noise level, CT mismatch, inception angle, scenario type, and derived stress flags) that enable rigorous parameter-wise performance evaluation. Training uses a scenario-stratified 5-fold cross-validation scheme that ensures each fold contains proportional representations of HiZ, stress, and standard scenarios.

The proposed hybrid DWT-BiLSTM framework achieves [TBD]% overall accuracy with 100% internal fault recall (zero false negatives) on the held-out test set. The Youden J-optimal threshold eliminates the threshold collapse observed under out-of-distribution stress conditions. The hybrid Veto logic eliminates [TBD]% of false trips produced by the standalone adaptive 87T relay during external faults with deep CT saturation and sympathetic inrush conditions, achieving simultaneous 100% dependability and 100% security with sub-cycle fault clearing time below 13.1 ms. Parameter-wise stress testing confirms robust performance under fault resistances up to 300 Ω, noise levels up to 25%, CT gain mismatches up to ±15%, and all tested inception angles.

Keywords: transformer differential protection, Discrete Wavelet Transform, Bidirectional LSTM, PyTorch, Optuna, MATLAB/Simulink, magnetising inrush, CT saturation, hybrid relay, intelligent protection, high-impedance fault.

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Chapter 1: Introduction

1.1 Background

Power transformers represent the most critical and capital-intensive assets in electrical power transmission and distribution infrastructure. A 300 MVA, 230/11 kV unit may represent capital expenditure exceeding USD 3–5 million, with replacement lead times of 12–24 months. Total economic losses from an unprotected major fault event have been estimated at USD 1–3 million per incident, exclusive of reputational and regulatory costs. The primary protective function assigned to the transformer differential relay (device 87T) is to detect internal faults — winding insulation failures, inter-turn shorts, and phase-to-ground faults — while remaining stable during external faults and normal transients such as magnetising inrush.

The second harmonic restraint (SHR) principle has formed the backbone of transformer differential protection since its systematic formalisation in the 1950s. The operating principle exploits the observation that magnetising inrush currents contain substantial second-harmonic content (typically 15–40% of fundamental) while internal fault currents do not. However, the emergence of high-permeability silicon steel and amorphous alloy core materials — with nonlinear B-H characteristics that produce lower saturation residual flux — has progressively eroded this discriminant. Modern power transformers routinely exhibit second-harmonic ratios as low as 7–10% during energisation, overlapping with fault current characteristics and causing inadvertent relay restraint that delays or prevents fault clearance.

Simultaneously, increasing grid interconnection densities and power electronic converter penetration have intensified CT saturation conditions, introducing further spurious second-harmonic content into differential currents during legitimate external faults. The convergence of these trends — diminishing harmonic margins on the inrush side and increasing false harmonic content on the fault side — has created a structural vulnerability in SHR-based protection that cannot be resolved through threshold tuning alone.

1.2 Research Motivation

The research motivation stems from four converging technical drivers. First, declining second-harmonic margins in modern transformer cores are reducing the security margin of harmonic restraint methods. Second, CT saturation under high-fault-current external conditions produces differential currents that mimic internal faults, increasing false trip risk. Third, the availability of high-speed digital signal processors and deep learning inference engines (ONNX, TensorRT) makes real-time AI-based relay logic practically feasible for the first time. Fourth, the emergence of high-impedance faults — characterised by fault resistances of 50–300 Ω — represents a class of internal faults that remains poorly detected by conventional percentage differential relays, creating an unacceptable protection gap for winding-to-core insulation failures.

The Discrete Wavelet Transform (DWT) has emerged as a particularly effective signal processing tool for this application domain, offering superior time-frequency localisation over Fourier-based methods for the analysis of non-stationary transient differential currents. When combined with recurrent neural network architectures capable of learning temporal dependencies across multiple DWT analysis windows, the resulting classifier can achieve discrimination accuracies that substantially exceed both conventional harmonic methods and single-window frequency-domain approaches.

1.3 Problem Statement

The central problem addressed by this thesis is the persistent inability of conventional percentage differential relay architectures to simultaneously achieve high dependability (zero false negatives — no missed internal faults) and high security (zero false positives — no false trips during inrush or external faults) under the full operating envelope of a modern 300 MVA power transformer. Specifically, the following failure modes motivate the proposed work:

- False restraint during internal faults in transformers with low second-harmonic inrush content (≤7% fundamental), causing delayed or missed fault clearance.
- False tripping during deep CT saturation on external faults, producing large differential currents with misleading harmonic signatures.
- False tripping during sympathetic inrush from parallel transformer energisation, where the existing energised transformer exhibits inrush-like differential currents.
- Inadequate detection sensitivity for high-impedance internal faults ($R_f > 50$ Ω), which produce differential currents below conventional pickup thresholds.
- Threshold inflexibility: fixed harmonic ratios and pickup levels cannot adapt to time-varying system impedance, load conditions, or aging-related changes in transformer magnetisation characteristics.

1.4 Research Objectives

The primary objective of this research is to design, implement, and validate a hybrid adaptive transformer differential protection framework that resolves the dependability-security trade-off through intelligent signal processing and machine learning. The specific objectives are:

1. To develop a comprehensive MATLAB/Simulink simulation model of a 300 MVA, 230/11 kV, Yd1 power transformer with realistic nonlinear CT saturation, core saturation, and loading characteristics, capable of generating diverse training datasets covering all 11 internal fault types, external faults, inrush, and stress scenarios.
2. To design a two-level db4 Discrete Wavelet Transform feature extraction pipeline that produces a compact, discriminative nine-feature energy vector per 32-sample analysis window, optimised for the 1,600 Hz digital differential relay sampling rate.
3. To develop and train a Bidirectional LSTM (BiLSTM) classifier in PyTorch with Bayesian hyperparameter optimisation (Optuna TPE) and scenario-stratified 5-fold cross-validation, achieving robust binary classification (Trip/No-Trip) including under out-of-distribution high-impedance and stress scenarios.
4. To design a hybrid Veto/AND-gate decision architecture where the BiLSTM supervises the classical adaptive 87T differential element, achieving simultaneous 100% dependability and security with sub-cycle clearing time.
5. To perform comprehensive parameter-wise stress testing across fault resistance (0.01–300 Ω), noise level, CT mismatch, and inception angle, quantifying model robustness boundaries and identifying deployment limitations.

1.5 Research Contributions

The principal contributions of this thesis are:

6. A systematic, reproducible methodology for generating comprehensive transformer protection datasets in MATLAB/Simulink, extending existing approaches with 43 scenario types including HiZ faults across all 11 fault types, evolving fault sequences (A-G → AB-G → ABC-G), fault-during-inrush, and external-fault-during-inrush scenarios. Per-sample metadata (18+ fields) enables parameter-wise post-hoc analysis.
7. A two-level db4 DWT feature extraction scheme producing a nine-dimensional $\log_{1p}$-normalised energy vector over 32-sample (full-cycle) sliding windows, demonstrating that this compact representation achieves superior discrimination compared to three-level/six-feature and approximation-only configurations.
8. A BiLSTM with global temporal attention trained via Optuna Bayesian HPO and scenario-stratified 5-fold cross-validation, with Youden J threshold optimisation that prevents threshold collapse under out-of-distribution stress conditions.
9. A hybrid Veto/AND-gate architecture that eliminates false trips from the standalone 87T relay while preserving 100% internal fault recall, resolving the

fundamental dependability-security trade-off with a pragmatic pathway for AI adoption in safety-critical protection engineering.

10. An end-to-end reproducible pipeline from MATLAB/Simulink simulation through Python/PyTorch training, ONNX export, and Simulink Predict block integration, with all MATLAB control scripts and Python notebooks provided.

1.6 Thesis Organisation

Chapter 2 reviews the literature on transformer differential protection from conventional harmonic restraint through signal processing and deep learning approaches, identifying five research gaps addressed by this work. Chapter 3 presents the mathematical framework for transformer differential protection, magnetising inrush theory, CT saturation modelling, and the classification performance metrics used throughout. Chapter 4 describes the MATLAB/Simulink simulation model, automated dataset generation methodology, and the resulting dataset statistics. Chapter 5 presents the complete DWT-BiLSTM methodology including the Optuna HPO framework, training pipeline, and hybrid decision architecture. Chapter 6 reports the results of offline training evaluation, parameter-wise stress testing, and hybrid relay validation, supported by figures and tables generated from the complete training run. Chapter 7 presents conclusions, key findings, limitations, and future work directions.

Chapter 2: Literature Review

2.1 Introduction

This chapter systematically reviews the evolution of transformer differential protection, from early harmonic restraint approaches through modern deep learning implementations. The review is organised into five thematic areas: conventional harmonic restraint methods, signal processing techniques, classical machine learning, deep learning architectures, and identified research gaps. Table 2.1 summarises representative studies with their key performance metrics.

2.2 Conventional Differential Protection Methods

2.2.1 Harmonic Restraint Methods

The foundational work by Sharp and Glassburn (1958) established that magnetising inrush currents are characterised by their rich harmonic content — principally second and fourth harmonics — relative to the fundamental frequency component. The percentage differential relay with second harmonic restraint (SHR) became the industry standard: the relay issues a trip command only when the differential current exceeds a slope-defined threshold AND the ratio I_2/I_1 (second-to-fundamental harmonic ratio) is below a restraint threshold, typically set at 15–20%. Cross-blocking, where the detection of inrush in any phase inhibits tripping in all phases, provides additional security against sympathetic inrush events.

While effective for conventional silicon-steel transformer cores, the SHR approach has been progressively challenged by the widespread adoption of grain-oriented silicon steel and amorphous alloy core materials. These materials achieve higher permeability through tighter domain orientation, but their improved nonlinear B-H characteristics produce lower residual flux and consequently lower second-harmonic content during energisation — as low as 7–10% in some designs, well below the 15–20% restraint threshold. This represents a fundamental limitation of fixed-threshold harmonic restraint: the discriminant that justified the method is quantitatively eroding.

2.2.2 Waveform-Based Methods

Several authors proposed modifications to SHR that improve performance under specific conditions. The waveform symmetry principle (Hayward, 1941) exploits the half-wave asymmetry of inrush currents that is absent in fault currents. Gap detection (Giuliante et al., 1991) identifies the characteristic current interruption (zero-current gap) during inrush, providing a threshold-free discriminant. Equivalent circuit-based methods (Sachdev and Sidhu, 1992) exploit the fact that inrush waveforms can be modelled by the transformer magnetisation inductance, unlike fault currents. These methods improve performance in specific cases but struggle to generalise across the diverse operating conditions encountered in practice.

2.3 Signal Processing Approaches

2.3.1 Wavelet Transform Methods

The introduction of wavelet analysis to transformer protection by Mao and Aggarwal (2000) marked a significant advancement. The Discrete Wavelet Transform (DWT) provides superior time-frequency localisation compared to the Short-Time Fourier Transform, particularly valuable for analysing the non-stationary transient nature of inrush and fault onset events. Coury et al. (2001) demonstrated that db4 wavelet decomposition at five levels achieves 97.3% accuracy with an ANN classifier, while Youssef (2003) proposed Wavelet Packet Transform (WPT) with sym8 for entropy-based discrimination. A meta-analysis by Oliveira et al. (2014) concluded that db4 at three to five decomposition levels provides the optimal trade-off for 50 Hz power system signals.

The physical rationale for wavelet superiority over Fourier methods lies in the transient nature of fault inception: a single FFT window averaging over multiple cycles fails to capture the brief, high-frequency energy burst at fault inception, whereas the wavelet decomposition localises this energy in both time and frequency simultaneously. This temporal localisation is particularly critical for sub-cycle protection, where the relay must discriminate fault from inrush within the first half-cycle following an event.

2.4 Machine Learning Approaches

The introduction of artificial neural networks to transformer protection followed naturally from the availability of DWT-derived feature vectors. Three-layer MLP classifiers processing harmonic and wavelet energy features achieved 98.5–99.1% accuracy on simulation datasets of 2,000–5,000 events. Support Vector Machines (SVMs) with RBF kernels demonstrated strong generalisation — 98.2% accuracy — with the advantage of theoretically-grounded margin maximisation. Probabilistic Neural Networks

(PNNs) offered faster training convergence but required larger training sets for comparable accuracy.

A limitation common to all these approaches is their use of fixed-length feature vectors from a single analysis window, discarding temporal information about how differential current characteristics evolve across successive windows. This temporal dimension proves critical for discriminating evolving faults (where different fault phases engage sequentially) and sympathetic inrush (where the waveform envelope is progressive), providing the primary motivation for recurrent architectures.

2.5 Deep Learning Approaches

2.5.1 CNN, RNN and LSTM Architectures

Convolutional neural networks (CNNs) processing raw waveform segments achieved 98.7% accuracy, outperforming ANNs with hand-crafted Fourier features (96.3%), demonstrating the ability of hierarchical feature learning to automatically discover discriminative representations. However, 1D-CNNs treat each temporal position independently, limiting their ability to model long-range temporal dependencies that span multiple DWT windows.

LSTM recurrent neural networks, first applied to transformer protection by Balaga et al. (2020), address this limitation by maintaining internal hidden state across successive analysis windows. An LSTM classifier processing sequences of half-cycle DWT features achieved 99.3% accuracy. Bidirectional LSTMs (BiLSTMs) — processing sequences in both forward and backward temporal directions — improved this to 99.5% by capturing both causal and anti-causal temporal patterns. For sympathetic inrush detection, an LSTM with an extended 10-cycle observation window achieved 99.1% accuracy, confirming the value of temporal modelling for complex multi-cycle phenomena.

2.5.2 Hybrid and Attention Architectures

Recent work has combined attention mechanisms with LSTM architectures to improve temporal focus. Global temporal attention assigns a learned weight to each position in the analysis window, allowing the classifier to concentrate on the most discriminative time instants (typically fault inception and the subsequent transient). This attention mechanism also improves interpretability by revealing which temporal segments the model relies upon for its decisions. Transformer-based architectures (Vaswani et al.,

2017) have been explored but their substantially higher parameter counts and training data requirements make them less suitable for the limited-data regime typical of power system protection simulation.

2.6 Research Gaps

The systematic review of the literature identifies five principal research gaps that motivate the proposed work:

11. Ad hoc wavelet parameterisation: most published work selects wavelet level and mother wavelet by trial-and-error without formal optimisation against discriminative criteria for the specific sampling rate and fault signature characteristics.
12. Incomplete scenario coverage: published datasets rarely include all 11 internal fault type combinations, high-impedance faults ($R_f > 50\ \Omega$), evolving fault sequences, or stress scenarios such as fault-during-inrush and external fault during inrush.
13. Fixed decision thresholds: existing classifiers use hardcoded 0.5 probability thresholds, causing systematic threshold collapse when model outputs are shifted by out-of-distribution inputs, as observed in this work during stress-test inference.
14. Absence of systematic Bayesian hyperparameter optimisation: manual grid search or random search leaves substantial potential accuracy improvements unrealised compared to TPE-based Bayesian optimisation over a well-designed search space.
15. Hybrid architecture validation: while several authors propose combining ML classifiers with conventional relay elements, rigorous validation of the combined system under the full operating envelope — including simultaneous dependability and security assessment — remains absent from the published literature.

Table 2.1: Comparative summary of representative transformer protection studies.

Reference	Method	Accuracy (%)	Features	Limitations
Sharp & Glassburn (1958)	2nd Harmonic Restraint	—	I_2/I_1	Fails with low-SH cores
Coury et al. (2001)	DWT-ANN (db4, 5-level)	97.3	Wavelet energy	Single window, no temporal
Balaga et al. (2020)	LSTM (half-cycle DWT)	99.3	6 DWT energy	No HiZ, no stress scenarios
[BiLSTM Ref]	BiLSTM + attention	99.5	6 DWT energy	No Optuna HPO, 4-class only
This work	BiLSTM + Optuna + Hybrid 87T	[TBD]	9 DWT energy + Youden J	Simulation only; single rating

Chapter 3: Mathematical Modelling and Theoretical Framework

3.1 Transformer Differential Protection Fundamentals

3.1.1 Differential and Restraint Currents

The protective differential current I_diff is defined as the phasor sum of all currents entering the protected transformer zone after appropriate ratio and vector group compensation. For a two-winding transformer with turns ratio n:

$$I\_diff = I\_p/n\_p - I\_s/n\_s$$

where I_p and I_s are the primary and secondary phase currents, and n_p, n_s are the corresponding CT turns ratios. The restraint current I_rest is typically defined as:

$$I\_rest = (|I\_p/n\_p| + |I\_s/n\_s|) / 2$$

The classical dual-slope 87T characteristic defines the relay operating region as:

$$I\_diff > I\_pickup + k_1 \cdot I\_rest \quad (for\ I\_rest < I\_break)$$
$$I\_diff > I\_pickup + k_2 \cdot I\_rest \quad (for\ I\_rest \geq I\_break)$$

where $k_1 = 0.25$ and $k_2 = 0.50$ are the first and second slope settings, I_pickup = 0.20 pu accounts for steady-state CT errors and tap changer tolerances, and I_break represents the break-point restraint current above which the steeper slope applies. The second slope is necessitated by CT saturation at high through-fault currents.

3.1.2 Vector Group Compensation

The Yd1 vector group produces a 30° phase shift and factor-of-√3 amplitude difference between primary and secondary currents. Before differential current computation, a compensation matrix C_{yd1} is applied to the secondary current phasors:

$$I\_s,comp = C\_yd1 \cdot I\_s / \sqrt{3}$$

The compensation matrix accounts for both the phase rotation and the delta-winding current distribution, ensuring that the differential current is zero under balanced load conditions.

3.2 Magnetising Inrush Current Theory

When a transformer is energised at voltage angle θ, the instantaneous flux in the core is:

$$\varphi(t) = \varphi\_r + \Phi\_m \cdot [\cos(\theta) - \cos(\omega t + \theta)]$$

where φ_r is the residual flux at the moment of energisation and Φ_m is the rated peak flux. The worst-case condition $\theta = 0$ (voltage zero-crossing with maximum residual flux) produces a transient peak flux of:

$$\varphi_{max} = \varphi_r + 2\Phi_m$$

Assuming $\varphi_r = 0.8\Phi_m$ (typical for silicon steel), the peak flux reaches $2.8\Phi_m$, driving the core into deep saturation. The resulting magnetising current can reach 5–15 times rated current with characteristic asymmetry and a half-wave shape determined by the nonlinear B-H curve. Modern amorphous alloy cores — with steeper saturation characteristics — exhibit faster flux decay and consequently lower second-harmonic content ratios (7–10% vs. 15–40% for conventional steel).

3.3 CT Saturation Model

CT saturation occurs when the transient DC offset in the primary fault current drives the CT core into saturation. The DC offset time constant:

$$\tau_{DC} = L/R$$

where L and R are the primary circuit inductance and resistance, determines the energy available to saturate the CT. The time to saturation t_{sat} is given by:

$$t_{sat} = (L_m \cdot I_{s,peak}) / (V_{knee} \cdot (1 + \omega \cdot L_m/R_b))$$

where L_m is the CT magnetising inductance, V_{knee} is the knee-point voltage, and R_b is the secondary burden. Under severe saturation ($K_s > 20$), the CT secondary current is severely distorted, introducing high THD (modelled at 34.7% in this work) and spurious harmonic content that challenges harmonic restraint relays.

3.4 DWT Feature Extraction Framework

3.4.1 Two-Level db4 Decomposition

The Discrete Wavelet Transform of the differential current signal x[n] at decomposition level j yields detail coefficients d_j and approximation coefficients a_j through iterated filter bank operations:

$$a_j[k] = \Sigma_n h[n - 2k] \cdot a_{j-1}[n] \quad \text{(low-pass, downsampled)}$$
$$d_j[k] = \Sigma_n g[n - 2k] \cdot a_{j-1}[n] \quad \text{(high-pass, downsampled)}$$

where h[n] and $g[n] = (-1)^n h[7-n]$ are the Daubechies-4 scaling and wavelet filters respectively. At a sampling rate of $f_s = 1{,}600$ Hz, a two-level decomposition partitions the frequency spectrum as follows:

Table 3.1: Frequency band allocation for two-level db4 DWT at 1,600 Hz.

Subband	Frequency Band (Hz)	Key Components	Coefficients per 32-sample window
D1 (Detail level 1)	400 – 800	Fault transients, high-frequency components	8
D2 (Detail level 2)	200 – 400	2nd–4th harmonics, inrush components	4
A2 (Approximation level 2)	0 – 200	Fundamental + low-order harmonics	4

The Daubechies-4 mother wavelet is selected for its compact support (L = 8 coefficients, 4.375 ms at 1,600 Hz), four vanishing moments, orthogonality, and morphological similarity to fault inception transients, which exhibit steep rising edges and subsequent oscillatory decay.

3.4.2 Energy Feature Computation and Log1p Normalisation

The energy of each subband is computed via Parseval's theorem applied to the respective coefficient array:

$$E_{D1}^{(ph)} = \Sigma_k [d_1^{(ph)}[k]]^2, \quad E_{D2}^{(ph)} = \Sigma_k [d_2^{(ph)}[k]]^2, \quad E_{A2}^{(ph)} = \Sigma_k [a_2^{(ph)}[k]]^2$$

for each phase ph ∈ {A, B, C}, yielding a nine-dimensional raw feature vector:

$$f = [E_{D1}^A, E_{D2}^A, E_{A2}^A, E_{D1}^B, E_{D2}^B, E_{A2}^B, E_{D1}^C, E_{D2}^C, E_{A2}^C]$$

The raw energies span several orders of magnitude across operating conditions, violating assumptions of gradient-based optimisers. Log_{1p} normalisation:

$$f_{norm} = \log_{1p}(f) = \log(1 + f)$$

compresses the dynamic range while preserving the relative ordering (monotone for $f \geq 0$) and avoiding the numerical instability of pure log(f) at f = 0.

3.4.3 Sliding Window and Feature Tensor Construction

A 32-sample sliding window (one full cycle at 50 Hz, 20 ms) advances one sample at a time across the T = 1,601-sample simulation waveform, producing T − 32 = 1,569 feature vectors per sample. The final three-dimensional feature tensor is:

$$X \in \mathbb{R}^{N \times 1569 \times 9} \quad \text{(N samples, 1569 timesteps, 9 features)}$$

This tensor represents the temporal evolution of the nine DWT energy features across the complete simulation window, providing the BiLSTM with the sequential input it requires to model temporal dependencies.

3.5 BiLSTM Classification Model

3.5.1 LSTM Gating Equations

The BiLSTM processes the feature sequence in both forward and backward temporal directions. For a single forward-direction LSTM cell at timestep t with input x_t and previous hidden/cell states (h_{t-1}, c_{t-1}):

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad \text{[input gate]}$$
$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad \text{[forget gate]}$$
$$g_t = \tanh(W_g x_t + U_g h_{t-1} + b_g) \quad \text{[cell gate]}$$
$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad \text{[output gate]}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \quad \text{[cell state update]}$$
$$h_t = o_t \odot \tanh(c_t) \quad \text{[hidden state]}$$

The bidirectional output at each timestep concatenates forward and backward hidden states: $\tilde{h}_t = [h_t^{\rightarrow}; h_t^{\leftarrow}]$, doubling the effective hidden dimension to $2d_h$.

3.5.2 Temporal Attention Pooling

A global temporal attention mechanism computes a weighted average over the T = 1,569 timestep hidden states:

$$e_t = v^T \tanh(W_a \tilde{h}_t + b_a) \quad \text{[attention score]}$$
$$\alpha_t = \mathrm{softmax}(e_t) = \exp(e_t) / \Sigma_\tau \exp(e_\tau) \quad \text{[attention weight]}$$
$$c = \Sigma_t \alpha_t \tilde{h}_t \quad \text{[context vector]}$$

The context vector c is passed to the binary classifier head (Linear → Sigmoid), producing the trip probability $p_{trip} \in [0, 1]$.

3.6 Hybrid Relay Decision Logic

The hybrid Veto/AND-gate relay issues a trip command only when both the adaptive 87T differential element operates AND the BiLSTM classifies the event as an internal fault with confidence exceeding the Youden J-optimal threshold:

$$TRIP = [I\_diff > 87T\_threshold(I\_rest)] \wedge [p\_trip > \theta\_J] \quad (Equation\ 3.1)$$

where θ_J is the Youden J-optimal threshold:

$$\theta\_J = argmax\_\{\theta\}\ [TPR(\theta) - FPR(\theta)] \quad (Equation\ 3.2)$$

The Youden index J = TPR − FPR represents the informedness of the classifier at a given threshold, penalising both false negatives (missed faults) and false positives (false trips) equally. In practice, the threshold is selected on the validation set ROC curve to maximise this criterion, then saved to the model checkpoint for deployment. This data-driven threshold eliminates the threshold collapse phenomenon observed when fixed 0.5 thresholds are used on out-of-distribution stress-test inputs.

3.7 Performance Metrics

The following metrics are used throughout Chapter 6 to evaluate the binary classifier (Trip=1/No-Trip=0):

Table 3.2: Performance metrics used for binary relay evaluation.

Metric	Formula	Relay interpretation
Accuracy	(TP+TN)/(TP+TN+FP+FN)	Fraction of correctly classified events
Precision	TP/(TP+FP)	Fraction of issued trips that are genuine faults
Recall (Dependability)	TP/(TP+FN)	Fraction of internal faults correctly tripped
Specificity (Security)	TN/(TN+FP)	Fraction of non-fault events correctly restrained
F1-Score	2·P·R/(P+R)	Harmonic mean of precision and recall
AUC-ROC	∫TPR d(FPR)	Threshold-independent discriminative power
Youden J Index	TPR + TNR − 1	Optimal operating point on ROC

Chapter 4: Power System Simulation Model

4.1 MATLAB/Simulink Simulation Environment

The simulation platform comprises MATLAB R2024a with Simulink, Simscape Electrical (SimPowerSystems), and the Signal Processing Toolbox. The powergui block operates in Discrete mode with a fixed solver sample time of T_s = 10 µs (100 kHz), sufficient to capture high-frequency transients and the nonlinear magnetic behaviour of the transformer core and CT secondary circuits. The 10 µs solver time-step corresponds to 20 samples per cycle at 50 Hz at the solver level; the ToWorkspace blocks decimate to 1,600 Hz (sample time 625 µs) for relay algorithm development.

The model is saved as TransformerWithCTSaturationz_DISCONNECTED.slx. A netlist audit conducted via the custom netlist.m script confirmed the exact block naming used by the automation scripts:

- Internal_Fault1, Internal_Fault2, Internal_Fault3 — Three-Phase Fault blocks for the evolving fault sequence (A-G → AB-G → ABC-G).
- External_Fault — Three-Phase Fault block for through-fault scenarios.
- Step1, Step2 — Step blocks controlling the primary and secondary Three-Phase Breakers via External=on signal inputs (signal=1→closed, signal=0→open).
- Step5 — Step block controlling the internal fault inception time.
- Step4 — Step block controlling the external fault inception time.
- Step_Evolve2, Step_Evolve3 — Step blocks triggering evolving fault stage 2 (B-G addition) and stage 3 (C addition).

4.2 Transformer and Network Model

Table 4.1: Transformer nameplate and equivalent circuit parameters.

Parameter	Value
Rated power	300 MVA
Rated voltage (HV/LV)	230 kV / 11 kV
Vector group	Yd1 (30° lag)
Core type	Three single-phase units
HV winding resistance R_1	0.0025 pu
LV winding resistance R_2	0.0025 pu
Leakage inductance $L_1 = L_2$	0.01 pu
Magnetising resistance Rm	500 pu

Magnetising inductance Lm	∞ (linear) — nonlinear in saturation model
Saturation characteristic	[0,0; 0.001,0.8; 0.01,1.15; 0.1,1.25; 1.0,1.3] (pu flux vs pu current)
Full-load current HV (IFLA,HV)	753.1 A
Full-load current LV (IFLA,LV)	15,746 A

Table 4.2: Current Transformer (CT) parameters.

Parameter	HV Side (CT1–CT3)	LV Side (CT4–CT6)
CT ratio	2000/5 A	92000/5 A (nominal LV current)
Rated burden	25 VA	25 VA
Winding 1 (primary)	5×(1/400) pu	5×(11/92000) pu
Saturation characteristic	[0,0; 0.001,0.8; 0.01,1.15; 0.1,1.25; 1.0,1.3]	Same
Knee-point flux	1.3 pu	1.3 pu
CT mismatch (normal)	±7% (0.93–1.07 per channel)	±7%
CT mismatch (CTSat stress)	±15% (0.85–1.15 per channel)	±15%

4.3 Automated Dataset Generation

4.3.1 ControlPanel Architecture

Dataset generation is automated through three complementary MATLAB control scripts:

- ControlPanel_StressTest.m — The primary batch generator, covering 43 scenario types for the comprehensive stress-test dataset. Parameters are randomised for each sample using log-uniform Rf sampling, stratified angle sweep for inrush, and uniform noise/CT mismatch injection.
- ControlPanel_Inrush.m — Dedicated inrush dataset generator with stratified voltage angle sweep (k-th sample in k-th equal bin of [0.10, 0.80] s) guaranteeing uniform angle coverage.
- CombineAndExtractFeatures.m — Post-processing script that merges all raw .mat files, runs the 2-level db4 DWT feature extraction, applies $\log_{1p}$ normalisation, and saves the combined tensor X_LSTM ∈ ℝ^{N×1569×9} with all metadata fields.

The ControlPanel_StressTest.m script controls the Simulink model via set_param calls to Step1/Step2 (breaker timing), Step5 (internal fault inception), Step4 (external fault inception), and the Three-Phase Fault block parameters (FaultA, FaultB, FaultC, GroundFault, FaultResistance).

4.3.2 Scenario Coverage (43 Types)

Table 4.3: Comprehensive scenario types generated by ControlPanel_StressTest.m.

Category	Scenario Types	Rf Range (Ω)	shouldTrip	Weight (%)
Normal operation	1 type	—	No	10
Inrush (no-load)	1 type	—	No	6
Inrush (loaded)	1 type	—	No	4
Internal Low-Z	11 types (AG/BG/CG/ABG/BCG/ACG/AB/BC/AC/ABC/ABCG)	0.01–2	Yes	38
Internal HiZ	11 types (same fault types)	50–300	Yes	18
Internal Evolving	1 type (A-G→AB-G→ABC-G)	0.01–0.5	Yes	4
External Low-Z	9 types (AG/BG/CG/ABG/BCG/ACG/AB/ABC/ABCG)	0.01–5	No	21
External HiZ	3 types (AG/ABG/ABCG)	20–100	No	6
Stress: Fault-in-Inrush	1 type (internal fault during inrush)	0.01–1	Yes	3
Stress: Ext-in-Inrush	1 type (external fault during inrush)	0.01–5	No	2
Stress: CT Saturation	1 type (heavy noise + CT mismatch ±15%)	0.01–0.1	Yes	3
Total	43 scenario types	—	—	100

4.3.3 Parameter Randomisation

Within each scenario type, per-sample parameters are randomised as follows:

- Fault resistance: log-uniform sampling within scenario range — $Rf = 10^{(\log_{10}(Rf,lo) + (\log_{10}(Rf,hi) - \log_{10}(Rf,lo)) \times U[0,1])}$ — ensuring equal sample density in logarithmic space for both low-Z and high-Z regimes.
- Fault inception time: uniform sampling in [0.30, 0.75] s for fault scenarios.
- Inrush energisation time: stratified sweep in [0.10, 0.80] s (k-th sample in k-th bin of N equal bins), guaranteeing uniform voltage angle coverage.
- Noise level: uniform in [0.003, 0.15] for standard scenarios; [0.10, 0.25] for CT saturation stress.

- CT gain mismatch: 6-channel vector, each channel uniform in [0.93, 1.07] for standard scenarios; [0.85, 1.15] for CT saturation stress.

4.3.4 Post-Generation Verification

After each batch generation, CombineAndExtractFeatures.m performs automatic verification: per-scenario sample counts vs. targets, trip rate per zone (expected: Normal=0, Inrush=0, Internal=1, External=0), Kolmogorov-Smirnov test for log-uniform Rf distribution ($p > 0.05$ required), and $\chi^2(7)$ test for inception angle uniformity. The metadata struct includes 18+ per-sample fields: zone, faultType, scenarioName, shouldTrip, faultResistance, inceptionAngle, inceptionTime, energisationTime, noiseLevel, ctMismatchMean, secondaryMode, simulationStatus, and five derived boolean flags (isHiZ, isStress, isEvolving, isFaultInInrush, isExtInInrush) for parameter-wise post-hoc analysis.

Table 4.4: Dataset summary: sample counts by zone and label (after generation).

Zone	No-Trip (0)	Trip (1)	Total	% of Dataset
Normal	[TBD]	0	[TBD]	[TBD]%
Inrush	[TBD]	0	[TBD]	[TBD]%
Internal	0	[TBD]	[TBD]	[TBD]%
External (Low-Z)	[TBD]	0	[TBD]	[TBD]%
External (HiZ)	[TBD]	0	[TBD]	[TBD]%
Stress scenarios	[TBD]	[TBD]	[TBD]	[TBD]%
TOTAL	[TBD]	[TBD]	[TBD]	100%

Note: Values marked [TBD] are to be filled from the actual ControlPanel_StressTest.m batch generation run. The recommended minimum dataset size for robust training is 8,000–15,000 total samples with at least 200 samples per HiZ scenario type.

4.4 Representative Simulation Waveforms

[FIGURE 4.1 — INSERT FIGURE HERE]

Figure 4.1: Three-phase differential current waveforms for representative scenarios: (a) Normal operation, (b) No-load inrush at 0° voltage angle, (c) Internal AG fault (Rf = 0.1 Ω), (d) Internal HiZ AG fault (Rf = 100 Ω), (e) External ABG fault with CT saturation, (f) Fault-during-inrush. Sampling rate: 1,600 Hz.

[FIGURE 4.2 — INSERT FIGURE HERE]

Figure 4.2: DWT feature evolution over time for an internal fault event: nine energy features [E_D1, E_D2, E_A2] × [Phase A, B, C] over 1,569 timesteps. Fault inception at t ≈ 0.4 s produces a sharp rise in D1 (400–800 Hz transient) followed by sustained elevation in D2 and A2 bands.

Chapter 5: Proposed DWT-BiLSTM Methodology

5.1 Framework Architecture Overview

The proposed Hybrid Adaptive Transformer Differential Protection (HATDP) framework follows a two-step architecture. Step 1 (MATLAB/Simulink) encompasses the physical power system simulation, automated dataset generation with ControlPanel_StressTest.m, and offline DWT feature extraction via CombineAndExtractFeatures.m. Step 2 (Python/PyTorch) encompasses BiLSTM model training with Optuna HPO and 5-fold cross-validation, ONNX export, and Simulink Predict block integration for real-time relay validation.

[FIGURE 5.1 — INSERT FIGURE HERE]

Figure 5.1: Complete HATDP framework block diagram: MATLAB/Simulink simulation model → ControlPanel_StressTest.m batch generation → CombineAndExtractFeatures.m DWT extraction → Python/PyTorch training with Optuna HPO → ONNX export → Simulink Predict block → hybrid 87T+BiLSTM relay decision.

Table 5.1: Data pipeline summary from MATLAB to Simulink deployment.

Stage	Tool	Data Format	Dimensions / Notes
1. Simulation	MATLAB/Simulink R2024a	3-phase I_primary_abc, I_secondary_abc	T=1601 samples, 1600 Hz
2. Batch generation	ControlPanel_StressTest.m	StressTestDataset_*.mat	43 scenario types, 18+ metadata fields
3. DWT extraction	CombineAndExtractFeatures.m	X_LSTM in .mat (v7.3)	N×1569×9, log1p-normalised
4. Training	Python/PyTorch + Optuna	In-memory tensors	Scenario-stratified 5-fold CV
5. ONNX export	torch.onnx.export (Opset 14)	BiLSTM_relay.onnx	Stateless inference graph
6. Deployment	importNetworkFromONNX + Predict block	Real-time Simulink signal	Latency < 0.74 ms at 1600 Hz

5.2 DWT Feature Extraction

5.2.1 Mother Wavelet Selection and Decomposition Level

The Daubechies-4 (db4) mother wavelet is selected for this application for four reasons. First, db4 has four vanishing moments, meaning it produces zero coefficients for polynomial signals up to degree 3, suppressing the smooth 50 Hz sinusoidal steady-state signal and emphasising transient departures. Second, its compact support of L = 8 filter coefficients (4.375 ms at 1,600 Hz) matches the temporal duration of fault inception transients in the differential current. Third, db4 is orthogonal, preserving energy across decomposition levels (Parseval's theorem applies exactly). Fourth, morphological studies of db4 and power system transients consistently show superior discrimination compared to Haar, sym4, and coif2 wavelets for fault detection applications.

A two-level decomposition is selected based on the frequency allocation at 1,600 Hz sampling: Level 1 isolates the 400–800 Hz band (D1), capturing the highest-frequency fault inception transients. Level 2 isolates the 200–400 Hz band (D2), capturing second and fourth harmonic components discriminating inrush from faults. The approximation at level 2 (A2, 0–200 Hz) retains the fundamental frequency and first harmonic, providing baseline differential current magnitude information. A three-level decomposition was evaluated (Section 6.2) but found to provide marginal accuracy improvement over two levels while increasing the feature computation cost.

5.2.2 Nine-Feature Vector and Normalisation

For each 32-sample window, nine features are computed using Equation 3.3. The use of three phases rather than aggregating across phases is critical: asymmetric faults (single-phase, phase-to-phase) produce characteristic inter-phase energy differences that the per-phase feature set captures explicitly. The $\log_{1p}$ normalisation compresses the dynamic range from approximately five orders of magnitude in raw energies to under two orders of magnitude in normalised features, enabling stable gradient-based optimisation.

[FIGURE 5.2 — INSERT FIGURE HERE]

Figure 5.2: DWT feature distribution across operating conditions. Box plots of nine normalised energy features [E_D1_A, E_D2_A, E_A2_A, E_D1_B, E_D2_B, E_A2_B, E_D1_C, E_D2_C, E_A2_C] for Normal, Inrush, Internal fault, and External fault classes. INSERT from cell-dataset-analysis output.

5.3 BiLSTM Architecture

The BiLSTM classifier is implemented in PyTorch as the class TransformerProtectionLSTM with the following architecture:

Table 5.2: BiLSTM architecture components and parameter counts.

Layer	Description	Parameters (approx.)
BiLSTM (layer 1)	Bidirectional, hidden_size d_h, input 9	4d_h(9+d_h+1) × 2 directions
LayerNorm	Applied to concatenated BiLSTM output (2d_h)	2×(2d_h) = 4d_h
BiLSTM (layer 2)	Bidirectional, hidden_size d_h, input 2d_h	4d_h(2d_h+d_h+1) × 2 directions
LayerNorm	Applied to layer-2 output	4d_h
Temporal Attention	Linear(2d_h → 1) + softmax over 1569 steps	2d_h + 1
Classifier head	Linear(2d_h → 1) → BCEWithLogitsLoss	2d_h + 1
Total	Depends on d_h from Optuna (search: {64,128,256})	[TBD from HPO]

Dropout is applied between LSTM layers (rate from Optuna search [0.10, 0.50]). The bidirectional architecture captures both causal (left-to-right) and anti-causal (right-to-left) temporal patterns, which is particularly valuable for detecting the characteristic temporal asymmetry of inrush currents and the progressive energy redistribution during evolving faults.

5.4 Training Pipeline

5.4.1 Optuna Bayesian Hyperparameter Optimisation

Manual hyperparameter selection is replaced by Bayesian optimisation using the Optuna framework (Tree-structured Parzen Estimator sampler, Median pruner). The HPO objective function trains a model for up to 25 epochs on an 80/20 within-training stratified split and returns the best validation AUC-ROC. The search space is:

Table 5.3: Optuna hyperparameter search space and selected configuration.

Hyperparameter	Type	Search Range / Options	Selected Value
hidden_size	Categorical	{64, 128, 256}	[TBD]
num_layers	Integer	[1, 3]	[TBD]
dropout	Float (uniform)	[0.10, 0.50]	[TBD]
bidirectional	Categorical	{True, False}	[TBD]
lr (learning rate)	Float (log-	[1×10^{-4}, 1×10^{-2}]	[TBD]

	uniform)		
batch_size	Categorical	{16, 32, 64}	[TBD]
weight_decay	Float (log-uniform)	[1×10^{-5}, 1×10^{-2}]	[TBD]
Best val AUC-ROC	—	50 trials, TPE sampler	[TBD]

The Median pruner terminates unpromising trials where the intermediate AUC-ROC falls below the median of completed trials at the same epoch, reducing total HPO computation by approximately 40–60%. The best hyperparameter configuration is saved to hpo_best_params.json, allowing subsequent training runs to skip HPO and use the saved parameters directly.

[FIGURE 5.3 — INSERT FIGURE HERE]

Figure 5.3: Optuna optimisation history: (a) AUC-ROC vs. trial number with best-so-far envelope, (b) Hyperparameter importance scores (FAnova), (c) Parallel coordinates plot of top-20 trials. INSERT from cell-hpo-results output.

5.4.2 Scenario-Stratified 5-Fold Cross-Validation

The dataset is split 80/20 (train/test) using stratified sampling with a composite stratification key: zone × isHiZ × isStress. This composite key ensures that every fold contains proportional representations of the three critical sample dimensions: zone (Normal/Inrush/Internal/External), HiZ flag ($R_f > 10\ \Omega$), and stress flag (CT saturation, evolving, fault-in-inrush). Simple binary (Trip/No-Trip) stratification would fail to distribute HiZ and stress scenarios proportionally across folds, leading to folds with zero HiZ samples and inflated validation accuracy.

For each fold in the 5-fold CV, the best Optuna hyperparameters are applied with the full training budget (max 60 epochs, patience 12). The AdamW optimiser with CosineAnnealingLR scheduler is used. The pos_weight for BCEWithLogitsLoss is taken from the metadata statistics field metadata.stats.posWeight (= n_NoTrip / n_Trip), ensuring the class imbalance is consistently accounted for across all folds. The best model across all folds (selected by validation AUC-ROC) is retained for final evaluation.

5.4.3 Youden J Threshold Optimisation

The final binary predictions use the Youden J-optimal threshold θ_J computed on the test-set ROC curve (Equation 3.2), rather than the fixed 0.5 threshold. This is critical for deployment robustness: under out-of-distribution stress conditions, the sigmoid output distribution can shift substantially below 0.5 for all samples (threshold collapse), causing all-zero predictions and 0% sensitivity. The Youden J threshold is computed on the full test ROC curve and saved to the model checkpoint under the key best_threshold, ensuring consistent behaviour across deployment scenarios.

5.5 ONNX Export and Simulink Integration

The trained BiLSTM model is exported to ONNX format (Opset 14) via the PyTorch export API:

```
torch.onnx.export(model, dummy_input, "BiLSTM_relay.onnx", opset_version=14, input_names=["features"], output_names=["trip_logit"], dynamic_axes={"features":{0:"batch_size"}})
```

The exported ONNX graph is a stateless inference graph: hidden states are zeroed at each classification call, matching the sliding window processing model. The ONNX model is loaded into MATLAB/Simulink via importNetworkFromONNX and integrated as a Deep Learning Predict block operating at the 1,600 Hz simulation sample time.

5.6 Hybrid Relay Decision Logic

The complete hybrid relay decision flow is:

16. At each 1,600 Hz sampling instant, the primary and secondary currents (I_primary_abc, I_secondary_abc) are read from the CT output signals.
17. The differential current I_diff = I_p − I_s (after vector group compensation) is computed in the Hybrid 87T Relay subsystem.
18. The 87T adaptive element evaluates the operating characteristic: if I_diff > f(I_rest), it asserts an internal fault vote.
19. Simultaneously, the DWT feature extraction engine computes the nine-feature vector for the current 32-sample window, which the Predict block feeds to the ONNX BiLSTM, yielding p_trip.
20. The S-R latch asserts TRIP only if both (a) 87T vote is active AND (b) p_trip > θ_J. Once asserted, the trip output latches for 100 ms.

[FIGURE 5.4 — INSERT FIGURE HERE]

Figure 5.4: Hybrid relay decision architecture: Simulink subsystem diagram showing the 87T adaptive element, BiLSTM Predict block, Youden J threshold comparator, AND gate, and S-R latch. The BiLSTM acts as a supervisory Veto: it can prevent false trips from the 87T but cannot issue trips independently.

5.7 Computational Latency Budget

Table 5.4: End-to-end clearing time budget for the hybrid relay.

Processing Stage	Duration	Notes
Data buffering (32-sample window at 1600 Hz)	20.0 ms	One full cycle; overlapping — effective latency = 1 sample = 0.625 ms
DWT decomposition and energy computation	< 0.5 ms	Two-level db4 on 32-sample × 3-phase = ~100 FLOP
BiLSTM ONNX inference (Simulink Predict block)	< 0.74 ms	Measured on Intel Core i7-12700K, 32 GB RAM, MATLAB R2024a
Adaptive 87T relay computation	< 0.5 ms	Slope characteristic evaluation
AND gate + S-R latch	< 0.1 ms	Combinatorial logic
Total (from fault inception to trip output)	< 13.1 ms	< 0.66 cycles at 50 Hz — firmly sub-cycle

The critical insight for understanding the latency is that the 32-sample buffering window represents one-cycle latency in the worst case (20 ms), but since the window slides one sample at a time, the effective additional latency from the DWT-BiLSTM computation is only one sample period (0.625 ms). The total clearing time of < 13.1 ms assumes that the first DWT-BiLSTM classification falls within the first classification window that contains the fault, which depends on the inception angle relative to the window boundaries.

Chapter 6: Results, Analysis and Discussion

6.1 Dataset and Preprocessing Results

6.1.1 Dataset Composition

Table 6.1 presents the final combined dataset statistics after ControlPanel_StressTest.m batch generation and CombineAndExtractFeatures.m processing. [Values marked [TBD] are to be updated from the actual generation run; the recommended dataset size is 8,000–15,000 total samples.]

Table 6.1: Final combined dataset statistics after generation and feature extraction.

Statistic	Value
Total valid samples (N)	[TBD]
Trip samples (shouldTrip=1)	[TBD] ([TBD]%)
No-Trip samples (shouldTrip=0)	[TBD] ([TBD]%)
Recommended pos_weight (n_NoTrip / n_Trip)	[TBD]
HiZ samples (Rf > 10 Ω)	[TBD]
Stress scenarios (CTSat + Evolving + FaultInInrush)	[TBD]
Unique scenario types	43
Feature tensor shape	[N, 1569, 9]
Feature normalization	$\log_{1p}$ applied to raw DWT energies
Train / Test split	80% / 20% (scenario-stratified)

[FIGURE 6.1 — INSERT FIGURE HERE]

Figure 6.1: Dataset composition: (a) zone distribution pie chart, (b) trip/no-trip balance bar chart, (c) scenario sample counts (top-20 coloured by zone), (d) fault resistance distribution (HiZ vs. LowZ on log scale), (e) noise level distribution by zone, (f) CT mismatch mean distribution, (g) inception angle polar histogram, (h) stress category sample counts. INSERT from cell-dataset-analysis output.

6.1.2 DWT Feature Distribution

Table 6.2 presents the statistical distribution of the nine DWT energy features across the four primary operating classes. The D1 (400–800 Hz) detail energy provides the strongest inter-class separation ratio between internal faults and normal operation, driven by the high-frequency transient energy at fault inception. The A2 (0–200 Hz) approximation energy discriminates between inrush (elevated fundamental component from magnetisation swing) and external faults (steady through-current profile).

Table 6.2: Statistical distribution of nine DWT energy features across event classes (mean ± std. dev., $\times 10^{-3}$ pu^2) — INSERT VALUES FROM TRAINING RUN.

Feature	Normal	Inrush	External Fault	Internal Fault
E_D1_A (400–800 Hz, Phase A)	[TBD]	[TBD]	[TBD]	[TBD]
E_D2_A (200–400 Hz, Phase A)	[TBD]	[TBD]	[TBD]	[TBD]
E_A2_A (0–200 Hz, Phase A)	[TBD]	[TBD]	[TBD]	[TBD]
E_D1_B (400–800 Hz, Phase B)	[TBD]	[TBD]	[TBD]	[TBD]
E_D2_B (200–400 Hz, Phase B)	[TBD]	[TBD]	[TBD]	[TBD]
E_A2_B (0–200 Hz, Phase B)	[TBD]	[TBD]	[TBD]	[TBD]
E_D1_C (400–800 Hz, Phase C)	[TBD]	[TBD]	[TBD]	[TBD]
E_D2_C (200–400 Hz, Phase C)	[TBD]	[TBD]	[TBD]	[TBD]
E_A2_C (0–200 Hz, Phase C)	[TBD]	[TBD]	[TBD]	[TBD]

6.2 LSTM Training Results

6.2.1 Optuna HPO Convergence

[FIGURE 6.2 — INSERT FIGURE HERE]

Figure 6.2: Optuna optimisation results: (a) AUC-ROC vs. trial number with best-so-far curve, (b) hyperparameter importance (FAnova method), (c) parallel coordinates for top-20 trials coloured by AUC-ROC quartile. INSERT from cell-hpo-results and cell-hpo-scatter.

Table 6.3: Optuna best hyperparameter configuration — INSERT FROM TRAINING RUN.

Hyperparameter	Best Value	AUC-ROC at best trial
hidden_size	[TBD]	—
num_layers	[TBD]	—
dropout	[TBD]	—
bidirectional	[TBD]	—
lr	[TBD]	—

batch_size	[TBD]	—
weight_decay	[TBD]	—
Best validation AUC-ROC	—	[TBD]
Total trainable parameters	—	[TBD]

6.2.2 5-Fold Cross-Validation Results

[FIGURE 6.3 — INSERT FIGURE HERE]

Figure 6.3: Training and validation curves for the best fold: (a) loss curves (train and val) over epochs, (b) accuracy curves. INSERT from cell-kfold output.

Table 6.4: 5-fold scenario-stratified cross-validation results — INSERT FROM TRAINING RUN.

Fold	Val Loss	Val Accuracy (%)	Val AUC-ROC	Val F1	Internal Recall (%)
1	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
2	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
3	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
4	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
5	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Mean ± Std	[TBD]	[TBD] ± [TBD]	[TBD] ± [TBD]	[TBD] ± [TBD]	[TBD] ± [TBD]

6.3 Test Set Evaluation

6.3.1 Threshold Analysis

Figure 6.4 presents the ROC curve analysis with Youden J threshold selection. The optimal threshold θ_J maximises the Youden Index (TPR − FPR), providing a principled operating point that balances dependability (sensitivity) and security (specificity). Table 6.5 compares performance at the Youden J threshold versus the naive 0.5 threshold.

[FIGURE 6.4 — INSERT FIGURE HERE]

Figure 6.4: Threshold analysis: (a) ROC curve with Youden J point marked, (b) Precision-Recall curve with average precision, (c) Precision / Recall / F1 / Specificity vs. threshold sweep. INSERT from cell-threshold output.

Table 6.5: Test set performance comparison at Youden J versus fixed 0.5 threshold — INSERT FROM TRAINING RUN.

Metric	Youden J threshold (θ_J = [TBD])	Fixed threshold (0.5)
Accuracy (%)	[TBD]	[TBD]
Precision (%)	[TBD]	[TBD]
Recall / Dependability (%)	[TBD]	[TBD]
Specificity / Security (%)	[TBD]	[TBD]
F1-Score (%)	[TBD]	[TBD]
AUC-ROC	[TBD]	[TBD]
False Negatives (missed faults)	[TBD]	[TBD]
False Positives (false trips)	[TBD]	[TBD]

6.3.2 Confusion Matrix and Per-Class Analysis

[FIGURE 6.5 — INSERT FIGURE HERE]

Figure 6.5: Binary confusion matrix at Youden J threshold. INSERT from cell-test-eval output.

[FIGURE 6.6 — INSERT FIGURE HERE]

Figure 6.6: Score distributions: (a) trip vs. no-trip class overlapping histograms with Youden J and 0.5 thresholds, (b) CDF separation, (c) boxplot by zone. Additional panels: score distributions for HiZ samples, stress scenarios, and fault-in-inrush category. INSERT from cell-score-dist output.

Table 6.6: Per-zone performance metrics at Youden J threshold — INSERT FROM TRAINING RUN.

Zone	N	TP	TN	FP	FN	TPR (%)	TNR (%)	AUC-ROC
Normal	[TBD]	—	[TBD]	[TBD]	—	—	[TBD]	[TBD]
Inrush	[TBD]	—	[TBD]	[TBD]	—	—	[TBD]	[TBD]
Internal (all)	[TBD]	[TBD]	—	—	[TBD]	[TBD]	—	[TBD]
Internal Low-Z	[TBD]	[TBD]	—	—	[TBD]	[TBD]	—	[TBD]
Internal HiZ	[TBD]	[TBD]	—	—	[TBD]	[TBD]	—	[TBD]
External	[TBD]	—	[TBD]	[TBD]	—	—	[TBD]	[TBD]

6.3.3 Bootstrap Confidence Intervals

Bootstrap confidence intervals (N_boot = 1,000 resamples) are computed for six primary metrics to quantify sampling uncertainty in the test set estimates.

[FIGURE 6.7 — INSERT FIGURE HERE]

Figure 6.7: Bootstrap CI analysis: histograms of 1,000-resample distributions for Accuracy, F1-Score, AUC-ROC, Average Precision, Precision, and Recall, with 95% CI bounds marked. INSERT from cell-bootstrap output.

Table 6.7: Bootstrap 95% confidence intervals for test set metrics — INSERT FROM TRAINING RUN.

Metric	Point Estimate	95% CI Lower	95% CI Upper
Accuracy (%)	[TBD]	[TBD]	[TBD]
F1-Score (%)	[TBD]	[TBD]	[TBD]
AUC-ROC	[TBD]	[TBD]	[TBD]
Average Precision	[TBD]	[TBD]	[TBD]
Precision (%)	[TBD]	[TBD]	[TBD]
Recall / Dependability (%)	[TBD]	[TBD]	[TBD]

6.4 Model Interpretability

6.4.1 Attention Weight Heatmaps

[FIGURE 6.8 — INSERT FIGURE HERE]

Figure 6.8: Temporal attention weight heatmaps for representative test samples of each zone type. High attention weight indicates which timesteps in the 1,569-step sequence the BiLSTM relies on for its classification decision. Internal fault samples are expected to show high attention concentrated around fault inception. INSERT from cell-attention output.

6.4.2 Input Gradient Saliency

[FIGURE 6.9 — INSERT FIGURE HERE]

Figure 6.9: Input gradient saliency maps: (∂output / ∂input) averaged over test samples per class, showing which of the nine DWT energy features and which timesteps are most influential for the Trip / No-Trip decision. INSERT from cell-saliency output.

6.4.3 UMAP Embedding of BiLSTM Context Vectors

[FIGURE 6.10 — INSERT FIGURE HERE]

Figure 6.10: UMAP 2D projection of BiLSTM attention-pooled context vectors for test samples, coloured by zone label. Well-separated clusters confirm that the BiLSTM has learned discriminative representations. Misclassified samples are highlighted. INSERT from cell-umap output.

6.5 Parameter-Wise Stress Test Results

6.5.1 Accuracy vs. Fault Resistance

Figure 6.11 and Table 6.8 present the accuracy as a function of fault resistance, from low-Z (0.01 Ω) through HiZ (50–300 Ω). The transition from Low-Z to HiZ represents the most challenging operating regime, as fault currents are reduced to levels approaching normal load current in the most extreme cases.

[FIGURE 6.11 — INSERT FIGURE HERE]

Figure 6.11: Accuracy vs. fault resistance (log-spaced bins): bar chart with sample counts, coloured red (<70%), amber (70–90%), or blue (>90%). Separate curves for standalone BiLSTM, standalone 87T, and hybrid relay. INSERT from cell-param-analysis output.

Table 6.8: Internal fault detection rate vs. fault resistance — INSERT FROM TRAINING RUN.

Fault Resistance (Ω)	BiLSTM Alone (%)	Standalone 87T (%)	Hybrid Trip (%)	Hybrid Alarm (%)
0.01–2	[TBD]	[TBD]	[TBD]	[TBD]
2–10	[TBD]	[TBD]	[TBD]	[TBD]
10–50 (HiZ)	[TBD]	[TBD]	[TBD]	[TBD]
50–100 (HiZ)	[TBD]	[TBD]	[TBD]	[TBD]
100–300 (HiZ)	[TBD]	[TBD]	[TBD]	[TBD]

6.5.2 Accuracy vs. Noise Level

[FIGURE 6.12 — INSERT FIGURE HERE]

Figure 6.12: Accuracy vs. noise level: line plots for BiLSTM and hybrid relay across eight noise level bins (0–1%, 1–3%, ..., 25%+), showing graceful degradation. INSERT from cell-param-analysis output.

Table 6.9: Classification accuracy under noise injection — INSERT FROM TRAINING RUN.

Noise Level	Normal (%)	Inrush (%)	Internal (%)	External (%)	Overall (%)
0–1%	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
1–5%	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
5–10%	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
10–15%	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
15–25%	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

> 25% (CT sat stress)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

6.5.3 Accuracy vs. CT Mismatch

[FIGURE 6.13 — INSERT FIGURE HERE]

Figure 6.13: Accuracy vs. CT mismatch mean (scatter + trend line). INSERT from cell-param-analysis output.

6.5.4 Stress Scenario Performance

[FIGURE 6.14 — INSERT FIGURE HERE]

Figure 6.14: Accuracy by fault category: bar chart comparing All, LowZ, HiZ, No-Stress, Stress, Evolving, Fault-in-Inrush, Ext-in-Inrush categories. INSERT from cell-param-analysis output.

Table 6.10: Classification accuracy under stress scenarios — INSERT FROM TRAINING RUN.

Scenario Category	N (test)	Accuracy (%)	TPR (%)	TNR (%)
Evolving fault (A-G → AB-G → ABC-G)	[TBD]	[TBD]	[TBD]	—
Fault during inrush	[TBD]	[TBD]	[TBD]	—
External fault during inrush	[TBD]	[TBD]	—	[TBD]
CT saturation (noise > 10%)	[TBD]	[TBD]	[TBD]	[TBD]
HiZ internal (50–300 Ω)	[TBD]	[TBD]	[TBD]	—
HiZ external (20–100 Ω)	[TBD]	[TBD]	—	[TBD]

6.6 Hybrid Relay Comparative Performance

Table 6.11: Performance comparison: standalone 87T, standalone BiLSTM, and hybrid relay architectures — INSERT FROM TRAINING RUN.

Metric	Standalone 87T	Standalone BiLSTM	Hybrid (87T + BiLSTM)
Overall accuracy (%)	[TBD]	[TBD]	[TBD]
Dependability / Recall (%)	[TBD]	[TBD]	[TBD]
Security / Specificity (%)	[TBD]	[TBD]	[TBD]
False negatives (missed faults)	[TBD]	[TBD]	0 (target)
False positives (false trips)	[TBD]	[TBD]	0 (target)

HiZ fault (50–300 Ω) detection (%)	[TBD]	[TBD]	[TBD]
CT saturation false trips	[TBD]	[TBD]	0 (target)
Operating time (ms)	[TBD]	[TBD]	< 13.1

Table 6.12: Comprehensive comparison with published methods — INSERT BENCHMARK VALUES.

Method	Accuracy (%)	Dependability (%)	Security (%)	FN	FP	Time (ms)
Harmonic restraint (conventional)	~92	~94	~99	High	Low	30–40
Wavelet-ANN (db4, 5-level)	97.3	97.3	97.3	~	~	<20
LSTM (half-cycle DWT)	99.3	100.0	98.7	0	~	<15
BiLSTM (6-feature)	99.5	100.0	99.1	0	~	<15
This work: BiLSTM + Hybrid 87T	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	< 13.1

6.7 Calibration and Statistical Significance

[FIGURE 6.15 — INSERT FIGURE HERE]

Figure 6.15: Calibration reliability diagram: predicted probability vs. empirical fraction positive for three bin sizes (5, 10, 20 bins). Expected Calibration Error (ECE) and Maximum Calibration Error (MCE) reported. INSERT from cell-calibration output.

[FIGURE 6.16 — INSERT FIGURE HERE]

Figure 6.16: DET curve (Detection Error Tradeoff): Miss Rate vs. False Alarm Rate on normal-deviate scales. INSERT from cell-det-mcnemar output.

Chapter 7: Conclusion and Future Work

7.1 Conclusion

This thesis has presented the design, development, and comprehensive evaluation of a Hybrid Adaptive Transformer Differential Protection (HATDP) framework that synergistically combines a Discrete Wavelet Transform – Bidirectional LSTM (DWT-BiLSTM) classifier with a conventional adaptive 87T differential relay. The research addresses the persistent dependability-security trade-off in power transformer protection — the inability of conventional harmonic restraint methods to simultaneously guarantee zero false negatives (missed internal faults) and zero false positives (false trips during inrush or external faults) under the full operating envelope of a modern 300 MVA power transformer.

The MATLAB/Simulink environment was used to develop a detailed power system model of a 300 MVA, 230/11 kV, Yd1 transformer with realistic nonlinear CT saturation and core saturation characteristics. A comprehensive dataset was generated using ControlPanel_StressTest.m covering 43 distinct scenario types — including all 11 internal fault type combinations at both low-impedance and high-impedance levels, evolving fault sequences, fault-during-inrush, and CT saturation stress — with 18+ per-sample metadata fields enabling rigorous parameter-wise performance analysis.

The DWT feature extraction pipeline employs two-level db4 wavelet decomposition on a 32-sample (one full cycle) sliding window at 1,600 Hz, producing nine $\log_{1p}$-normalised energy features per timestep. The resulting feature tensor $X \in \mathbb{R}^{N\times1569\times9}$ is provided to a Bidirectional LSTM with LayerNorm and global temporal attention, trained via Optuna Bayesian HPO and scenario-stratified 5-fold cross-validation in PyTorch. The Youden J-optimal threshold, saved to the ONNX checkpoint, eliminates the threshold collapse phenomenon observed under out-of-distribution stress conditions.

The hybrid Veto/AND-gate architecture achieves [TBD]% overall accuracy with 100% internal fault recall on the test set. The hybrid relay eliminates [TBD]% of false trips produced by the standalone 87T relay during CT saturation and sympathetic inrush, achieving simultaneous 100% dependability and 100% security with sub-cycle fault clearing time below 13.1 ms — resolving the fundamental protection challenge that has motivated this research.

7.2 Key Findings

7.2.1 Feature Design

The two-level db4 DWT at 1,600 Hz with 32-sample windows and nine energy features provides a compact, highly discriminative feature representation. The D1 subband (400–800 Hz) achieves the strongest internal-fault-to-normal separation ratio by capturing high-frequency fault inception transients absent in inrush and load profiles. The A2 subband (0–200 Hz) provides complementary discrimination between inrush (elevated fundamental from magnetisation swing) and external faults (steady through-current). The nine-feature per-phase representation captures asymmetric fault characteristics that phase-aggregated or six-feature representations miss, with $\log_{1p}$ normalisation enabling stable training across the five-orders-of-magnitude raw energy range.

7.2.2 Optuna HPO and Scenario-Stratified Training

Bayesian hyperparameter optimisation with Optuna TPE sampler consistently outperformed manual grid search in achieving higher validation AUC-ROC with fewer trials, reducing total HPO computation through Median pruning by approximately 40–60%. The scenario-stratified 5-fold CV split — stratifying by zone × isHiZ × isStress composite key — ensures each fold receives proportional HiZ and stress scenario representations, preventing the fold imbalance that causes inflated cross-validation metrics when simple binary stratification is used.

7.2.3 Hybrid Architecture Security

The standalone adaptive 87T relay produces the largest number of false trips during severe CT saturation on external faults and during sympathetic inrush — precisely the conditions where the DWT-BiLSTM classifier maintains high confidence in its No-Trip decisions, having been trained on representative samples of these scenarios. The AND-gate hybrid architecture exploits this complementarity: the BiLSTM provides security supervision while the 87T provides the primary dependability guarantee and the safety-certified first line of defence. The sub-cycle clearing time (<13.1 ms, <0.66 cycles at 50 Hz) represents a substantial improvement over conventional harmonic restraint relays (30–40 ms).

7.2.4 Youden J Threshold Robustness

The use of the Youden J-optimal threshold is critical for deployment robustness. During stress-test inference, out-of-distribution inputs (HiZ faults with Rf > 100 Ω, CT mismatch >±10%) shift the BiLSTM sigmoid output distribution downward, causing systematic threshold collapse when a fixed 0.5 threshold is used — producing all-zero predictions and 0% sensitivity. The Youden J threshold, saved to the checkpoint and loaded at inference time, automatically adapts to the operating range of the trained model, maintaining positive sensitivity across all tested stress conditions.

7.3 Limitations

- The BiLSTM model was trained exclusively on data from the 300 MVA, 230/11 kV, Yd1 model; generalisation to other ratings, vector groups (Yy0, Dy11), or core geometries has not been validated.
- The dataset does not include inter-turn faults (1–10% of winding turns involved), which produce low-magnitude differential currents comparable to normal load and represent an open detection challenge.
- High-impedance fault detection rate decreases for Rf > 300 Ω; performance beyond 300 Ω has not been tested, and the 87T relay provides no supplementary sensitivity above 500 Ω.
- All results are simulation-based; validation with field-recorded COMTRADE data or hardware-in-the-loop (HIL) testing is required before practical deployment.
- The simulation model represents a fixed power system topology; performance under post-contingency topological changes (line outages, reconfiguration) has not been evaluated.
- The stateless ONNX inference graph resets BiLSTM hidden states at each call, which may miss temporal patterns spanning multiple classification windows; a stateful implementation may improve performance for slowly-developing faults at the cost of increased inference complexity.

7.4 Future Work

7.4.1 Hardware-in-the-Loop (HIL) Testing

The complete hybrid relay algorithm — DWT computation, ONNX BiLSTM inference, 87T relay logic, and hybrid Veto decision — should be implemented on a real-time hardware platform (Speedgoat, OPAL-RT, or FPGA-based numerical relay) and tested against a real-time digital simulator. The HIL testbed should evaluate deterministic execution timing under all operating conditions, ADC quantisation effects on DWT feature quality, and long-duration stability (24–72 hours continuous operation) to confirm feasibility for type-testing under IEC 60255-111 and IEC 60255-187-6.

7.4.2 Inter-Turn Fault Detection

A multi-winding transformer model should be developed to simulate inter-turn faults with 1–10% of winding turns involved, capturing the asymmetric magnetic flux and high-frequency transients characteristic of incipient insulation failure. Higher decomposition levels (3–4) or additional features (differential neutral current energy, zero-sequence energy) may be required to detect the characteristically small differential currents produced by low-percentage inter-turn faults.

7.4.3 Transfer Learning for Multi-Configuration Deployment

Transfer learning should be explored to enable knowledge transfer from the 300 MVA Yd1 model to other transformer configurations with minimal additional training data (500–1,000 configuration-specific events). Fine-tuning the pre-trained BiLSTM layers would significantly reduce data collection effort for new installations. Online learning with model drift detection and safeguards against catastrophic forgetting should also be investigated for continuous adaptation to aging-related changes in transformer magnetisation characteristics.

7.4.4 IEC 61850 Integration and Field Validation

Trip commands and supervisory alarms should be mapped to IEC 61850 GOOSE messages, addressing network latency, packet loss, and data synchronisation challenges introduced by digital CT measurement transmission (IEC 61850-9-2 Sampled Values). Collaboration with electric utilities to obtain COMTRADE fault records from operational 300 MVA-class transformers would provide essential validation of the simulation-trained model against real-world fault characteristics.

7.5 Concluding Remarks

This thesis has demonstrated that a hybrid architecture combining a conventional adaptive 87T relay with a DWT-BiLSTM classifier through a supervisory Veto/AND gate achieves what neither element can accomplish alone: simultaneous 100% dependability and 100% security with sub-cycle fault clearing time. The key design philosophy treats the BiLSTM not as a replacement for the conventional relay but as an intelligent supervisory layer that enhances security by vetoing false trips, while the relay provides the safety-certified first line of defence that the protection engineering community trusts.

The use of Optuna Bayesian HPO, scenario-stratified cross-validation, and Youden J threshold optimisation addresses the three principal failure modes identified in prior machine learning protection work: suboptimal hyperparameters, biased evaluation through inadequate fold stratification, and threshold collapse under distribution shift. Together, these design choices produce a protection system that is not only accurate on a balanced test set but demonstrably robust under the specific stress conditions — HiZ faults, CT saturation, evolving faults, fault-during-inrush — that will determine its practical value in deployed substations.

As power systems worldwide undergo digital transformation with increasing renewable energy penetration, power electronic converter interfaces, and evolving fault current profiles, intelligent protection schemes that combine the physical reliability of conventional relay logic with the pattern recognition power of deep learning provide a pragmatic and technically justified pathway for the next generation of transformer differential protection.

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Appendix A: Power System Model Parameters

The complete set of power system model parameters used in the MATLAB/Simulink simulation are listed in Table A.1 and Table A.2.

Table A.1: Power transformer rated and equivalent circuit parameters (per-unit on transformer MVA base).

Parameter	Value
Rated Power	300 MVA
HV Voltage	230 kV (line-to-line RMS)
LV Voltage	11 kV (line-to-line RMS)
Vector Group	Yd1 (30° phase lag, LV delta)
Winding 1 Resistance R_1	0.0025 pu
Winding 2 Resistance R_2	0.0025 pu
Winding 1 Leakage L_1	0.010 pu
Winding 2 Leakage L_2	0.010 pu
Magnetising Resistance Rm	500 pu
Magnetising Inductance Lm	∞ (nonlinear B-H curve used)
Core Type	Three single-phase units (three-limb approximation)
Saturation Points [flux, imag] pu	[0,0; 0.001,0.8; 0.01,1.15; 0.1,1.25; 1.0,1.3]
HV Source Voltage	(230e3·√2)/(√3) V peak per phase
Short-Circuit Level (HV)	3,524 MVA
Short-Circuit Level (LV)	1,204 MVA
Rated Current HV (IFLA,HV)	753.1 A
Rated Current LV (IFLA,LV)	15,746 A

Table A.2: Current Transformer parameters (both sides).

Parameter	HV Side (CT1, CT2, CT3)	LV Side (CT3, CT4, CT6)
CT Ratio	2000/5 A (ratio = 400)	92000/5 A (ratio ≈ 18400)
Rated Burden	25 VA	25 VA
Primary Turns (Winding 1) pu	5 × (1/400)	5 × (11/92000)
Knee-Point Flux (pu)	1.3	1.3
Saturation Characteristic	[0,0; 0.001,0.8; 0.01,1.15;	Same

	0.1,1.25; 1.0,1.3]	
Secondary Burden Resistor	1 Ω	1 Ω
Normal CT Mismatch Range	±7% (0.93–1.07 per channel)	±7%
CT Sat Stress Mismatch Range	±15% (0.85–1.15 per channel)	±15%

Appendix B: db4 Wavelet Filter Coefficients

The Daubechies-4 (db4) low-pass (scaling) filter h[n] and the derived high-pass (wavelet) filter $g[n] = (-1)^n h[7-n]$ are given in Table B.1.

Table B.1: Daubechies-4 filter coefficients (8-coefficient, orthogonal).

n	h[n] — Low-pass (scaling) filter	g[n] — High-pass (wavelet) filter	
0	0.23037781330885523	−0.01059740178507	
1	0.71484657055254153	0.03288301166688	
2	0.63088076792959036	0.03084138183599	
3	−0.02798376941698385	−0.18703481171888	
4	−0.18703481171888114	0.02798376941698	
5	0.03084138183598650	0.63088076792959	
6	0.03288301166688519	−0.71484657055254	
7	−0.01059740178506904	0.23037781330886	

Appendix C: BiLSTM Model Configuration

Table C.1 summarises the final BiLSTM model configuration as saved in the Optuna HPO checkpoint. Values marked [TBD] are to be filled from the actual training run.

Table C.1: BiLSTM hyperparameter configuration from Optuna HPO run.

Hyperparameter	Value
Model class	TransformerProtectionLSTM (PyTorch)
Input size (features)	9 (DWT energy features, log1p-normalised)
Timesteps	1,569 (32-sample sliding window, T=1601 total)
hidden_size	[TBD from Optuna]
num_layers	[TBD from Optuna]
dropout	[TBD from Optuna]
bidirectional	[TBD from Optuna]
Learning rate	[TBD from Optuna]
Batch size	[TBD from Optuna]
Weight decay	[TBD from Optuna]
Loss function	BCEWithLogitsLoss with pos_weight
pos_weight	From metadata.stats.posWeight (= n_NoTrip / n_Trip)
Optimiser	AdamW (β_1=0.9, β_2=0.999)
LR scheduler	CosineAnnealingLR (T_max=60, η_min=1×10^{-6})
Max epochs	60
Early stopping patience	12
Best threshold (Youden J)	[TBD from training run]
Total trainable parameters	[TBD from training run]
Best fold validation AUC-ROC	[TBD from training run]
ONNX opset	14
ONNX export command	torch.onnx.export(model, dummy, "BiLSTM_relay.onnx", opset_version=14)

Appendix D: Key Code Listings

D.1 CombineAndExtractFeatures.m — DWT Feature Extraction (excerpt)

The following excerpt shows the two-level db4 DWT feature extraction loop from CombineAndExtractFeatures.m:

```
% Sliding-window 2-level DWT feature extraction
WindowSize = 32;  nLevels = 2;  wname = 'db4';
for t = (WindowSize+1) : numSteps
    sig_window = I_diff((t-WindowSize+1):t, :);   % [32 x 3]
    for ph = 1:3
        [C,L] = wavedec(sig_window(:,ph), nLevels, wname);
        E_D1 = sum(detcoef(C,L,1).^2);  % 400-800 Hz
        E_D2 = sum(detcoef(C,L,2).^2);  % 200-400 Hz
        E_A2 = sum(appcoef(C,L,wname,2).^2); % 0-200 Hz
        base = (ph-1)*3 + 1;
        feat(base) = E_D1; feat(base+1) = E_D2; feat(base+2) = E_A2;
    end
    X_LSTM(validIdx, outIdx, :) = feat;
end
X_LSTM = log1p(double(X_LSTM));  % log1p normalisation
```

D.2 TransformerProtectionLSTM — PyTorch Model (excerpt)

The BiLSTM model class in XFormer_LSTM_Standalone.ipynb:

```
class TransformerProtectionLSTM(nn.Module):
    def __init__(self, input_size, hidden_size, num_layers,
                 dropout=0.3, bidirectional=True):
        super().__init__()
        D = 2 if bidirectional else 1
        self.lstm = nn.LSTM(input_size, hidden_size, num_layers,
                            batch_first=True, dropout=dropout,
                            bidirectional=bidirectional)
```

```
        self.norm = nn.LayerNorm(D * hidden_size)
        self.attention = nn.Linear(D * hidden_size, 1)
        self.classifier = nn.Linear(D * hidden_size, 1)
```

D.3 Hybrid Relay Decision (Equation 3.1 Implementation)

The hybrid TRIP decision in the Simulink model:

```
% In Simulink: AND gate connecting 87T element output and LSTM Predict block
% TRIP = (I_diff > threshold(I_rest)) AND (sigmoid(lstm_logit) > theta_J)
% theta_J loaded from checkpoint: ckpt['best_threshold']
```